GITARTHA GOGOI

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Embedded Firmware Engineer · Real-Time Control · STM32 / ESP32 / FreeRTOS

Education

M.Tech, Digital Communication & Signal Processing — Sikkim Manipal Institute of Technology CGPA 8.6/10 · 2024–2026
Thesis: adaptive fertigation controller on STM32L4 / FreeRTOS

B.Tech, Electronics & Communication Engineering — Assam University CGPA 7.6/10 · 2020–2024

Experience

Embedded Firmware & Hardware Design Engineer, 4Climate Jan 2026 – Present
Industrial inline batch fertigation controller · STM32L471 (Cortex-M4) + ESP32-S3 · FreeRTOS · C

- Developed firmware for a **dual-MCU controller**: the **STM32L471** runs the real-time dosing/irrigation **FSM**, actuation, persistence and scheduling as **FreeRTOS** tasks over message queues (replacing a super-loop); the **ESP32-S3** runs the **LVGL** UI and **AWS IoT** MQTT sync over 9 **Device Shadows**, linked by a framed binary **UART** protocol.
- Designed a **two-degree-of-freedom EC/pH dosing controller**: **PI** feedback trim on a **feedforward** term from a MATLAB tank-mixing model, ported to fixed-step discretized C (**model-based design**), with pulse dosing and deadband/hysteresis against valve chatter; holds concentration within ±**3%** of setpoint in steady state.
- Root-caused a systematic ∼**67% EC overdose** after refills: the learned plant gain $k_{est} = S/V$ (stock strength / tank volume) was learned at full volume but applied at the refill-trigger volume. Designed a **mass-ledger** replacement that tracks nutrient mass and volume ($EC = M/V$) and demotes the probe to slow drift correction.
- Implemented an **adaptive exponential filter** (gain scaled by measured signal variance) on EC/pH (**I2C**) and ultrasonic level streams, with range, rate-of-change and stuck-value checks rejecting faulted readings.
- Implemented **power-cut recovery** for zone-sequenced irrigation: a CRC-protected **RTC backup-register** checkpoint (event, zone, remaining time, sequence counter) drives a safe-state → refill → resume-at-zone restart.
- Diagnosed field **MCU resets** with a 3-channel **oscilloscope**: isolated 8–17.5 V common-mode **ground bounce** from inductive AC loads, ruled out relay coils and cable length, traced the likely path to a UART-error `NVIC_SystemReset()`, and specified snubbers, flyback diodes and single-point grounding.
- Ran an 11–14-day, two-board **bench endurance test** (∼**175k pulses**) via a custom harness with CRC-protected, wear-throttled **EEPROM** counters read live over **SWD**; exposed a pump-housing failure *within* normal dose limits and 25–37.5% flow loss.
- Automated guided **two-point EC/pH calibration** with coefficients in non-volatile storage, cutting manual calibration effort by ∼**40%**; measured pump delivery at ∼76% of nameplate and derived piecewise **flow-calibration** curves.

Team Lead – Embedded Control, Eminence Robotics Feb 2022 – Aug 2022
Competition robotics · ESP32 · C++

- Tightened ESP32 actuator **control-loop timing** (interrupt-driven acquisition, timer-scheduled loop execution), improving actuator response time by **18%**; added startup self-checks and runtime fault flags to cut fault-driven downtime.
- Tuned wireless telemetry for RF-noisy arenas and led the team to **3 national-level** robotics competition wins.

IoT Systems Intern, MeOWL Technologies Sep 2021 – Dec 2021
Distributed industrial telemetry · ESP32 · Raspberry Pi CM4

- Built a multi-node acquisition pipeline (ESP32 edge nodes → Raspberry Pi CM4 aggregator) and root-caused **Wi-Fi reconnection/backoff** failures in deployed ESP32 units, improving field uptime by **30%**.
- Engineered a **LoRa/GSM failover** that switches transport on primary-link loss, eliminating single-link telemetry outages.

Technical Skills

Languages: C, C++, Python, MATLAB | **Tools:** MATLAB/Simulink, Git/GitHub, VS Code, Docker, LaTeX, Linux
MCU / RTOS: STM32L4 (HAL), ESP32 / ESP32-S3 (ESP-IDF, LVGL), Raspberry Pi CM4; FreeRTOS, bare-metal
Protocols & Cloud: UART, I2C, SPI, LoRa, GSM, Wi-Fi, MQTT, AWS IoT Core (Device Shadow)
Control & Signal: feedforward/PI control, adaptive gain learning, pulse/deadband dosing, adaptive filtering, sensor validation, model-based design
Debug & Test: oscilloscope, ST-LINK/SWD live watch, STM32CubeIDE, bench endurance testing, EMI/ground-noise diagnosis

Research, Publications & Service

- **IRS-Assisted MU-MISO Downlink** (M.Tech research, MATLAB): **alternating optimization** of IRS phase shifts with **WMMSE precoding** (active IRS: **22%** gain over passive); **power-aware active/passive element allocation** for hybrid IRS, whose EE advantage over passive holds in the large-array regime ($N \approx 512$).
- G. Gogoi, T. Bose, P. Vishwakarma, S. N. Sur, "Performance Analysis of Active IRS-Assisted MU-MISO System," *Int. J. Inf. Technol.*, vol. 17, pp. 5135–5146, 2025.
- G. Gogoi *et al.*, "Optimizing Energy and Spectral Efficiency in Hybrid IRS-Assisted MU-MISO Systems through Power-Aware Element Allocation," *Physical Communication* (Elsevier), 2026.
- Peer Reviewer, International Conference on Communication, Devices and Networking (ICCDN), 2025–26.